

SAI SURYA VAMSI TUMPATI

+1(512) 999-3586 ◊ Austin, TX

suryatumpati@gmail.com ◊ [linkedin.com/suryatumpati](https://www.linkedin.com/suryatumpati) ◊ [Github.com/suryatumpati](https://github.com/suryatumpati)

SUMMARY

Computer Science graduate and AI developer with experience designing and deploying GenAI tools, NLP pipelines, and full-stack systems. Skilled in Python, Docker, and building scalable language model-based microservices using LangChain, Hugging Face, and Pinecone. Proven ability to deploy end-to-end ML pipelines on AWS infrastructure.

SKILLS

Skills: Python, SQL, Docker, AWS, CI/CD, Git, LangChain, OpenAI API, Pinecone, RAG, LLMs, Hugging Face

Tools & Concepts: Prompt Engineering, JWT Auth, Terraform, Streamlit, Selenium, Regression testing.

EDUCATION

Bachelor of Computer Science

2022–2025

University of North Alabama, Florence, AL

PROJECTS

ThinkSync – Real-Time Collaboration Platform

Tech Stack: Django, Python, React.js, HTML, CSS, AWS EC2, S3

- Deployed backend on **AWS EC2** with static/media on **S3**, improving performance by **30%**. Set up **CI/CD** via GitHub Actions on a Linux server.
- Achieved **40% faster page loads** through asset delivery optimization and improved front-end rendering.
- Integrated **RESTful APIs** to enable real-time data synchronization and support for concurrent users. ([Code on GitHub](#))

HealthLine Assistant – AI-Powered Medical Knowledge Tool

Tech Stack: Python, LangChain, OpenAI API, Pinecone, Hugging Face

- Built an assistant providing instant, accurate answers from a **100K+** document semantic index, improving medical search clarity by **80%**
- Developed a retrieval-augmented generation (RAG) pipeline combining semantic search and OpenAI's LLMs, reducing irrelevant responses by 65%
- Cut user response time to 1.5s per query via optimized Pinecone indexing and asynchronous backend
- Improved verified textbook sources which reduces misinformation risk. ([Code on GitHub](#))

Automated Finance Dashboard – Real-Time Monitoring Platform

Tech Stack: Python, Streamlit, Pandas

- Designed and built a real-time finance dashboard to monitor over **1,000+ monthly transactions** with trend visualizations. Conducted regression testing after each update.
- Automated ETL pipeline and processing of raw data from CSV file, reducing manual analysis time by **90%**.
- Integrated logging and exception handling to ensure the reliability of **99.9% script**, supporting daily insights.
- Deployed via an EC2 instance for **24/7 availability**, enabling mobile-friendly visualization. ([Code on GitHub](#))

Pathway – AI-Driven Job Search and Recruitment Platform

Tech Stack: React, Node.js, Express, MongoDB, OpenAI API

- Developed a full-stack job platform that supports **10K+ listings**, featuring intelligent filtering mechanisms and role-based access control for recruiters and candidates.
- Integrated OpenAI API to generate job descriptions, reducing the manual workload of recruiters by **60%**.
- Automated the back-end infrastructure with secure JWT-based authentication and optimized MongoDB data flows, ensuring a consistent **99.95% uptime**.

CERTIFICATES AND LICENSES

- **AWS Certified Cloud Practitioner** — Amazon Web Services, March 2025
Skills Gained: Cloud Fundamentals, AWS core services, pricing models, and security principles.